COMMERCIAL FRAMEWORK SPECIFICATION

VVU HBK Mk-II

Terminal-as-a-Service (TaaS)

Defining the commercial architecture for municipal water infrastructure intelligence through a non-recourse OpEx subscription model, 100% equity retention, and sole data sovereignty. This specification establishes the financial, governance, and operational pillars that govern the HBK Mk-II Hydro-Gateway deployment lifecycle.

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1. Service Definition — Terminal-as-a-Service (TaaS)

1.1 The TaaS Value Proposition

The Terminal-as-a-Service (TaaS) model represents a fundamental strategic pivot from traditional CapEx-based hardware procurement toward a Non-Recourse OpEx Treatment framework for municipal water infrastructure. Under conventional procurement paradigms, municipalities are compelled to allocate substantial upfront capital for hardware acquisition, installation, and commissioning—creating an immediate balance sheet liability with no guarantee of operational performance or data yield. The TaaS framework eliminates this structural inefficiency by transferring the entire hardware lifecycle—from manufacturing through deployment, maintenance, and eventual decommissioning—under VVU's operational umbrella.

The key value propositions of the TaaS model are threefold. First, **Balance Sheet De-risking**: municipalities receive the HBK Mk-II Hydro-Gateway Assembly as a fully operational service unit without assuming ownership liability. The hardware remains on VVU's balance sheet, not the client's, ensuring that depreciation risk, obsolescence exposure, and residual value uncertainty are entirely absorbed by VVU. Second, **Elimination of Municipal Upfront Capital Constraints**: the subscription-based OpEx structure allows municipalities to deploy advanced leak-localization infrastructure within existing operational budgets, bypassing the often-prohibitive CapEx approval cycles that delay critical infrastructure modernization by years. Third, **Performance-Linked Alignment**: because VVU retains 100% equity and sole data sovereignty, the commercial incentive is directly coupled to operational excellence—VVU profits only when the terminal delivers measurable infrastructure intelligence.

1.2 Asset Identification — HBK Mk-II / Hydro-Gateway Assembly

The HBK Mk-II Hydro-Gateway Assembly comprises 11 integrated components, each positioned within a precise XYZ coordinate framework that defines the physical architecture of the terminal. This assembly represents a complete, self-contained water infrastructure intelligence unit—from primary flow measurement at the inlet to secure telemetry transmission at the mast apex. The coordinate system establishes the spatial relationships necessary for structural integrity, hydraulic coupling, and Bayesian inference processing. Each component serves a defined functional role within the Observation Vector (O_t) generation pipeline.

Component	X (mm)	Y (mm)	Z (mm)	Functional Role
Pressure Pipe	0	0	750	Nodal Head Estimation
South Datum Skid	0	-560	40	Base structural support
North Datum Skid	0	+560	40	Base structural support
Left Service Rack	-720	-240	290	Lateral mounting
Right Service Rack	+720	+240	290	Lateral mounting
Edge Control Cabinet	0	+400	400	Local probabilistic processing / Bayesian inference
Power Backup Module	0	-400	400	Power redundancy

Component	X (mm)	Y (mm)	Z (mm)	Functional Role
Inlet Meter Pod	-750	-160	930	Primary flow measurement
Outlet Meter Pod	+750	+160	930	Secondary flow measurement
Telemetry Mast	0	0	1290	Secure data transmission
Top Height Beacon	+100	0	1465	Physical site identification / Aviation safety

The coordinate architecture ensures that each component's spatial placement optimizes for signal propagation, structural load distribution, and hydraulic flow continuity. The datum skids (North and South) anchor the assembly to the installation surface, while the service racks provide lateral stability. The Edge Control Cabinet houses the Bayesian inference engine that transforms raw sensor observations into probabilistic leak-localization outputs. The dual meter pods (Inlet and Outlet) generate the differential flow measurements essential for the Observation Vector, and the Telemetry Mast transmits encrypted data packets to VVU's central processing infrastructure.

2. Core Commercial Pillars

2.1 100% Equity Retention

VVU maintains absolute ownership of every HBK Mk-II Hydro-Gateway Assembly deployed under the TaaS framework. This is not a lease arrangement with transfer provisions, nor a financed purchase with a vesting schedule—it is an uncompromised equity position that persists from manufacturing through deployment, operation, and eventual decommissioning or redeployment. The 100% equity retention principle serves three critical commercial functions.

Non-Transferable Rights: No client, partner, or third party may acquire ownership stakes in the deployed hardware. This ensures that VVU's commercial relationship with the municipality remains a pure service contract—the client pays for intelligence delivery, not asset acquisition. The hardware never becomes a liability on the client's balance sheet, and VVU never dilutes its control over the terminal's operational parameters.

Asset Security: Because VVU retains full ownership, the hardware cannot be seized, lien-attached, or otherwise encumbered by municipal financial distress. In the event of client default or contract termination, VVU exercises its unilateral recovery right to physically extract and redeploy the assembly at a new service location. This structural protection transforms the hardware from a sunk-cost risk into a mobile, redeployable commercial asset.

Capital Efficiency Rights: The 100% equity position grants VVU the exclusive right to retrieve and redeploy HBK Mk-II assets across successive service engagements. A terminal deployed at Municipality A can, upon contract conclusion, be recovered, refurbished, and redeployed at Municipality B—transforming each hardware unit from a single-deployment cost center into a multi-lifecycle revenue generator. This capital efficiency model dramatically reduces per-deployment amortization and accelerates the path to positive unit economics.

2.2 Sole Data Sovereignty

Under the TaaS framework, VVU holds exclusive and non-shareable sovereignty over all data generated by the deployed HBK Mk-II terminal. The Observation Vector (O_t) comprises four primary measurement dimensions: **P** (Pressure), **F** (Flow), **A** (Acoustic signature), and **T** (Temperature). These four dimensions constitute the raw intelligence feed from which VVU's Bayesian inference engine derives probabilistic leak-localization outputs. No client, academic partner, or third-party vendor may access, replicate, or redistribute this data without VVU's explicit written authorization.

The sovereignty framework is enforced through **Data-Audit Inseparability**: every Observation Vector packet is cryptographically bound to its generating terminal via SHA-256 hash chaining. This creates an immutable audit trail that ensures data provenance, integrity, and non-tampering. The hash chain links each data point to its specific terminal identity, deployment timestamp, and sensor calibration state—making it impossible to detach data from its originating context. Furthermore, **IP Protection** is inherent in the sovereignty model: the Bayesian inference algorithms, probabilistic models, and localization methodologies that transform raw O_t data into actionable intelligence are VVU's proprietary intellectual property. The data sovereignty framework ensures that these IP assets cannot be reverse-engineered or exfiltrated through data access.

2.3 InfrastructureRight Abstraction

The TaaS framework introduces the InfrastructureRight Abstraction—a four-pillar resource interface that transforms physical infrastructure access into digital rights within a pooled resource interface. The four pillars are: **Water** (hydraulic access and flow measurement rights), **Energy** (power supply and backup entitlements), **Compute** (Bayesian inference processing allocation), and **Storage** (telemetry data retention and archival capacity). Each pillar is defined as a quantifiable, contractually guaranteed service level that the municipality receives as part of its TaaS subscription.

This abstraction serves a critical architectural purpose: it decouples the client's infrastructure requirements from the specific hardware implementation. The municipality does not purchase "a pressure sensor" or "a flow meter"—it subscribes to a Water Right that guarantees a defined level of hydraulic intelligence delivery. If VVU upgrades the sensor array, replaces a component, or reconfigures the inference pipeline, the client's service level remains constant—the InfrastructureRight is hardware-agnostic. This design ensures that VVU can continuously improve the terminal's capabilities without triggering contract renegotiations or client-facing configuration changes.

3. Financial Model & Revenue Distribution

3.1 Subscription-Based OpEx Structure

The TaaS financial model is built on a recurring operational expense (OpEx) subscription structure that replaces the traditional capital expenditure (CapEx) procurement cycle. Under this model, the municipality pays a fixed monthly or quarterly subscription fee that covers three bundled service components: **Hardware Leasing** (the physical HBK Mk-II assembly deployment and maintenance), **Bayesian Inference Services** (the probabilistic leak-localization processing that transforms raw sensor data into actionable intelligence), and **Operational Maintenance** (ongoing calibration, component replacement, and firmware updates). The subscription fee is structured as a fixed operational cost with defined escalation caps, ensuring budget predictability for the municipality across the contract lifecycle.

This OpEx structure delivers immediate commercial advantages: the municipality avoids the large upfront capital outlay that typically requires multi-year budget approval cycles, VVU gains predictable recurring revenue that supports long-term R&D; investment, and the performance-linked nature of the subscription ensures that VVU's revenue is contingent on delivering measurable infrastructure intelligence rather than merely deploying hardware. The subscription model also enables flexible scaling—municipalities can adjust their InfrastructureRight allocations (additional Compute, expanded Storage, enhanced Water measurement density) within the subscription framework without renegotiating the core contract.

3.2 The 60/30/10 Revenue Split

All TaaS subscription revenue is distributed according to a fixed 60/30/10 allocation model that balances operational sustainability, growth investment, and risk governance. This split is contractually fixed and cannot be modified without triparty agreement (VVU, Academic Leads, Municipal Client) as defined in Section 4.1.

Allocation	Percentage	Scope	Description
Operational Baseline	60%	Hardware + Operations	Hardware amortization, server costs, HBK inference engine maintenance, field service operations
Growth & R&D;	30%	Software + Research	Software refinement, Bayesian algorithm improvement, academic collaboration funding, next-generation terminal development
Risk & Compliance	10%	Audit + Regulatory	ProofBridge-Liner audit chain operations, regulatory alignment, SRS parameter verification, compliance documentation

3.3 Incentive Alignment

The 30% Growth & R&D; allocation is not merely a cost center—it is the incentive alignment mechanism that ensures continuous improvement in the terminal's leak-localization accuracy. Under the TaaS SLA framework (Section 5), VVU commits to achieving a search radius of 500 meters or less for leak localization. The 30% R&D; allocation funds the Bayesian algorithm refinement, academic research partnerships, and next-generation sensor development that continuously narrow the search radius toward this target. This creates a self-reinforcing commercial cycle: improved accuracy increases client satisfaction and contract renewal rates, which in turn generates additional subscription revenue that funds further R&D; investment. The 60/30/10 split ensures that this cycle is structurally financed—not dependent on ad hoc budget allocations or external grant funding.

4. Risk Mitigation & "Zero Fabrication" Governance

4.1 Administrative Shielding — Triparty SRS Delegation

The TaaS governance framework implements a Triparty SRS (System Requirements Specification) Delegation Addendum known as the "Three Keys" mechanism. Under this governance structure, no single party can unilaterally modify, waive, or bypass any SRS parameter. The three keyholders are: **VVU** (the service provider and hardware owner), **Academic Leads** (the independent research validators), and **Municipal Client** (the service subscriber and infrastructure operator). Any modification to the SRS requires simultaneous authorization from all three keyholders, ensuring that commercial pressures from VVU, research ambitions from academic partners, or operational convenience from the municipality cannot individually compromise the terminal's safety and performance parameters.

This administrative shielding is particularly critical in the context of the Zero Fabrication Mandate (Section 4.2). The Three Keys mechanism ensures that no manufacturing funds can be released, no SRS parameter can be relaxed, and no deployment can proceed without triparty consensus. The governance structure transforms the SRS from a unilateral specification into a multilateral contract—protecting all parties from the consequences of premature or under-specified hardware production.

4.2 Zero Fabrication Mandate

The Zero Fabrication Mandate is the foundational risk governance principle of the TaaS framework: no manufacturing funds shall be released until all 32 SRS parameters have been verified and validated. This mandate exists to prevent the most catastrophic failure mode in infrastructure hardware production—manufacturing a terminal that has not been fully specified, tested, and certified. The 32 SRS parameters encompass every dimension of the terminal's operational specification: hydraulic performance, structural integrity, electromagnetic compatibility, environmental resilience, and Bayesian inference accuracy.

Parameter	Boundary Value	Unit	Governance Scope
Surge Maximum	20	bar	Hydraulic pressure safety ceiling — no deployment above this threshold
Wave Celerity	1400	m/s	Acoustic propagation speed limit for Bayesian inference calibration
Corrosion Limit	10.00	mm	Maximum permissible material degradation before mandatory component replacement
Structural Bound (FoS)	1.5 - 3.0	ratio	Factor of Safety range for structural load-bearing components under all operating conditions

4.3 Verification Gates VR1-VR5

The Zero Fabrication Mandate is enforced through a five-stage verification gate system that progressively validates the terminal from geometric specification through field acceptance. Each gate represents a mandatory checkpoint that must be passed before the next stage can proceed. No gate

may be skipped, combined, or conditionally waived. The verification gates establish a linear, non-parallel validation path that ensures every dimension of the terminal's specification is independently confirmed.

Gate	Designation	Scope	Validation Criteria
VR1	Geometry	Physical dimensions and spatial layout	All 11 component XYZ positions verified against SRS coordinate specification
VR2	Material	Material composition and properties	Material certificates, corrosion resistance testing, and structural strength validation
VR3	Assembly	Component integration and coupling	Inter-component connectivity, hydraulic seal integrity, and electrical bonding verification
VR4	Functional (FAT)	Factory Acceptance Testing	Full functional testing under simulated operational conditions, Bayesian inference output validation
VR5	Field Acceptance (SAT)	Site Acceptance Testing	In-situ performance validation at deployment site, live hydraulic conditions, telemetry confirmation

5. Operational Performance Targets (SLA Metrics)

5.1 Leak Localization Accuracy

The primary SLA metric for the TaaS framework is leak localization accuracy, measured as the reduction in search uncertainty relative to conventional detection methods. The HBK Mk-II terminal, operating under its Bayesian inference engine, commits to achieving a **95% reduction in search uncertainty** compared to traditional acoustic correlation methods. This translates to a **maximum search radius of 500 meters** for any detected leak event. The search radius metric is the most operationally significant performance indicator: it defines the physical area that municipal repair teams must inspect to locate the leak source. A 500-meter search radius means that excavation and repair resources can be deployed with precision rather than dispersed across kilometers of pipeline infrastructure.

The accuracy target is achieved through the convergence of three technical capabilities: the Observation Vector's four-dimensional sensor data (P, F, A, T), the Bayesian inference engine's probabilistic spatial mapping, and the SHA-256 hash-chained data integrity that ensures no sensor drift or data corruption degrades the localization output. The 95% reduction target is not a static benchmark—it is a continuously improving metric driven by the 30% R&D; revenue allocation that funds ongoing algorithm refinement and sensor calibration research.

5.2 False-Positive Suppression

False-positive events represent the most costly failure mode in leak detection operations: each false alarm triggers unnecessary excavation, pipeline shutdown, and resource deployment that wastes municipal operational budget and erodes confidence in the detection system. The TaaS SLA framework

addresses this through a strict **False-Positive Rate (FPR) target of 5% or less**, enforced via Poisson-Gaussian mixture distributions in the Bayesian inference engine. The Poisson component models the discrete event frequency (how many leak events occur in a given observation window), while the Gaussian component models the continuous measurement noise (sensor signal variability under normal operating conditions). The mixture distribution enables the inference engine to distinguish between genuine hydraulic anomalies (actual leaks) and statistical noise fluctuations (false triggers) with high confidence.

The 5% FPR target is validated against historical pipeline data from comparable municipal networks and is subject to continuous recalibration as the terminal accumulates site-specific operational data. Each false-positive event is logged, analyzed, and fed back into the inference engine's training dataset—ensuring that the FPR metric improves over the deployment lifecycle rather than degrading.

5.3 Information Density Guarantee

The Information Density (ID) metric is the ultimate efficiency indicator for the TaaS framework, defined as: **ID = Information Gained / Deployment Cost**. This ratio measures the operational intelligence yield per unit of financial investment. Unlike raw accuracy metrics that measure detection performance in isolation, Information Density contextualizes performance within the commercial framework—ensuring that the intelligence delivery is not merely accurate but economically efficient.

The ID metric serves two governance functions. First, it provides the Municipal Client with a quantifiable return-on-investment indicator: each subscription payment can be evaluated against the volume and precision of infrastructure intelligence received. Second, it provides VVU with an internal efficiency benchmark: if the ID ratio declines (indicating that deployment costs are rising faster than intelligence yield), the 30% R&D; allocation is directed toward the specific technical bottlenecks—sensor calibration, algorithm refinement, or hardware optimization—that are degrading the ratio.

SLA Metric	Target	Measurement Method	Governance Mechanism
Leak Localization Accuracy	95% reduction / 500m radius	Bayesian inference spatial probability mapping	30% R&D; allocation for continuous algorithm refinement
False-Positive Rate	5% or less	Poisson-Gaussian mixture distribution analysis	Continuous recalibration against site-specific operational data
Information Density	ID = Info Gained / Deploy Cost	Ratio of intelligence yield to subscription investment	Internal efficiency benchmark driving R&D; priority allocation

6. Vendor Financing Terms

6.1 Tooling Deposit — Tranche 1 Budget Lock

The TaaS vendor financing framework establishes a structured capital release mechanism that enforces the Zero Fabrication Mandate at the financial level. The **Tranche 1 Budget Lock** of **R812,490.00** is a ring-fenced tooling deposit that covers the production mold fabrication and the KCL (Kinematic Calibration Library) parametric library development. This deposit is released only upon successful completion of Verification Gate VR1 (Geometry) and VR2 (Material)—ensuring that no manufacturing capital is deployed before the terminal's physical and material specifications have been independently validated by the triparty keyholder structure.

The Tranche 1 deposit structure serves two protective functions. First, it ensures that VVU's manufacturing investment is preceded by validated specifications—preventing the costly scenario of producing molds against unverified or incomplete SRS parameters. Second, it provides the fabrication partner with a committed capital baseline that enables production planning without financial uncertainty. The Budget Lock is non-revocable once released: VVU cannot retract the tooling deposit after VR1/VR2 completion, providing the fabrication partner with contractual assurance that their production investment will be honored.

6.2 Prototyping Exemptions — Limited Experimental Validation Builds

While the Zero Fabrication Mandate prohibits production manufacturing before full SRS verification, the framework recognizes the necessity of limited experimental validation builds for testing and calibration purposes. The **Prototyping Exemptions** clause permits fractional, ring-fenced capital releases for Limited Experimental Validation Builds—small-scale production runs (typically 1-3 units) intended solely for laboratory testing, Bayesian inference calibration, and field simulation exercises. These exemption builds are strictly governed: they may not be deployed at client sites, they may not be sold or transferred, and they must be decommissioned after their validation lifecycle concludes.

The prototyping exemption framework ensures that VVU can validate the terminal's inference accuracy, sensor calibration, and hydraulic performance under controlled conditions without violating the Zero Fabrication principle. The exemption builds are funded from the 30% R&D; allocation—not from the Tranche 1 tooling deposit—ensuring that production capital and validation capital are financially separated. This separation prevents the common failure mode where prototyping costs erode production budgets, forcing manufacturing compromises that degrade the final product.

6.3 Default & Asset Recovery

The default and asset recovery framework is the ultimate enforcement mechanism of VVU's 100% equity retention principle. In the event of client default—whether through subscription payment failure, contract violation, or operational non-compliance—VVU exercises its **unilateral right to recover and redeploy HBK Mk-II assets**. This right is absolute and non-negotiable: no client agreement, municipal regulation, or third-party claim may prevent VVU from physically extracting the deployed terminal and relocating it to a new service engagement.

The recovery framework operates under a defined procedural sequence. Upon default event confirmation, VVU issues a formal Recovery Notice to the Municipal Client with a 30-day compliance window. If the default is not remediated within this window, VVU dispatches a field recovery team to decommission and extract the terminal assembly. The recovered asset enters VVU's refurbishment pipeline: components are inspected, recalibrated, and upgraded as necessary before redeployment at a subsequent service location. The recovery cost is absorbed by VVU's 10% Risk & Compliance allocation—ensuring that asset recovery does not impose additional financial burden on the Operational Baseline or R&D; budget lines.

This recovery mechanism transforms the TaaS model from a rental arrangement into a true service lifecycle: the hardware is never "sold and forgotten" but remains a perpetually managed commercial asset that generates revenue across multiple deployment cycles. The unilateral recovery right ensures that VVU's capital investment in each HBK Mk-II unit is protected regardless of individual client outcomes—de-risking the overall portfolio and enabling confident investment in next-generation terminal development.

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